

LAB EXPERIMENT RECORD

Set 10: Survey Responses Analysis

Department	AIML
Course Code / Course Name	DSA0601 / Data Handling and Visualization
Experiment	Data Visualization using R and Tableau
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Aim

To analyze survey response data using a bar chart, stacked bar chart, and tabular representation, and to develop a Tableau dashboard combining the visualizations for interactive exploration.

Dataset: Survey Responses

Survey ID	Question 1	Question 2	Question 3
1	A	B	C
2	B	A	D
3	C	A	B

PROGRAM 1: Grouped Bar Chart for Question 1

Create a grouped bar chart to visualize the distribution of answers for Question 1. Label the chart elements.

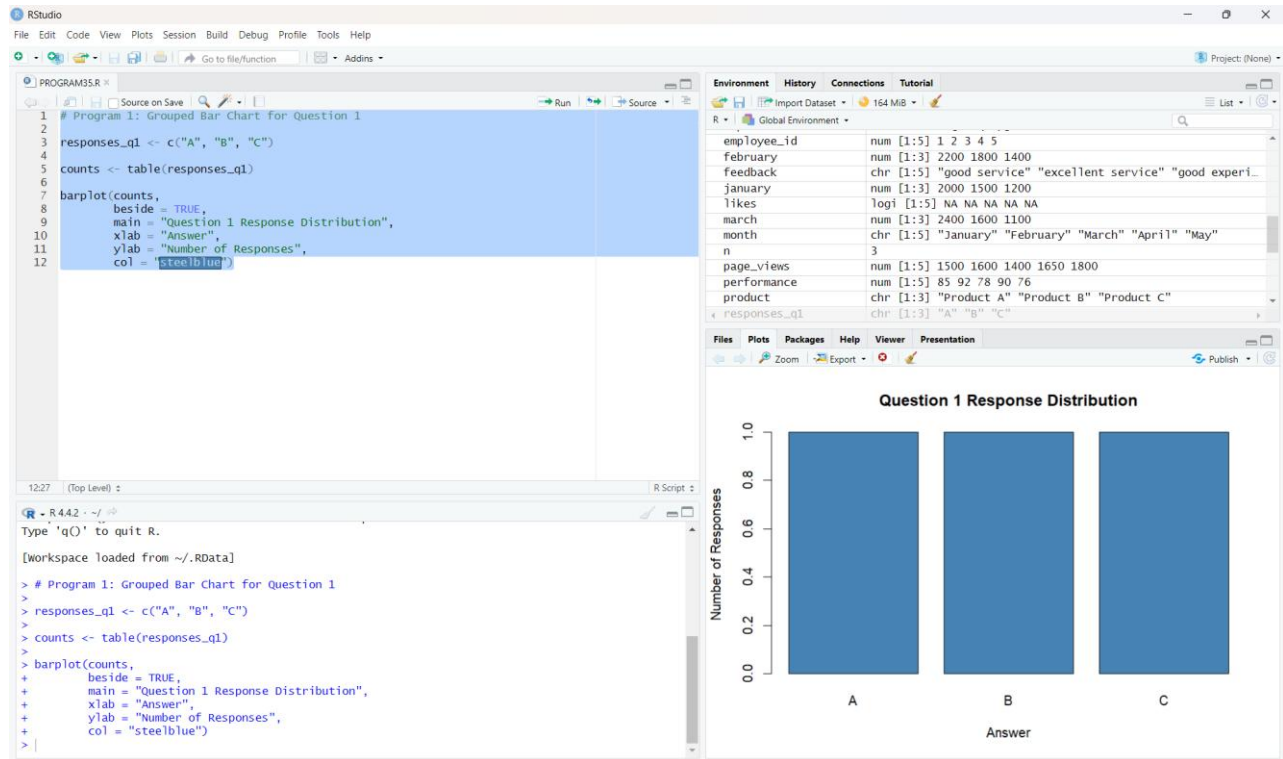
```
# Program 1: Grouped Bar Chart for Question 1

responses_q1 <- c("A", "B", "C")

counts <- table(responses_q1)

barplot(counts,
        beside = TRUE,
        main = "Question 1 Response Distribution",
        xlab = "Answer",
        ylab = "Number of Responses",
        col = "steelblue")
```

Output 1: Question 1 Response Distribution



Output 1: Grouped Bar Chart for Question 1

Observation: The chart shows that answers A, B, and C each occur once in the given survey responses.

PROGRAM 2: Stacked Bar Chart for All Questions

Generate a stacked bar chart to represent the overall distribution of responses for all three questions.

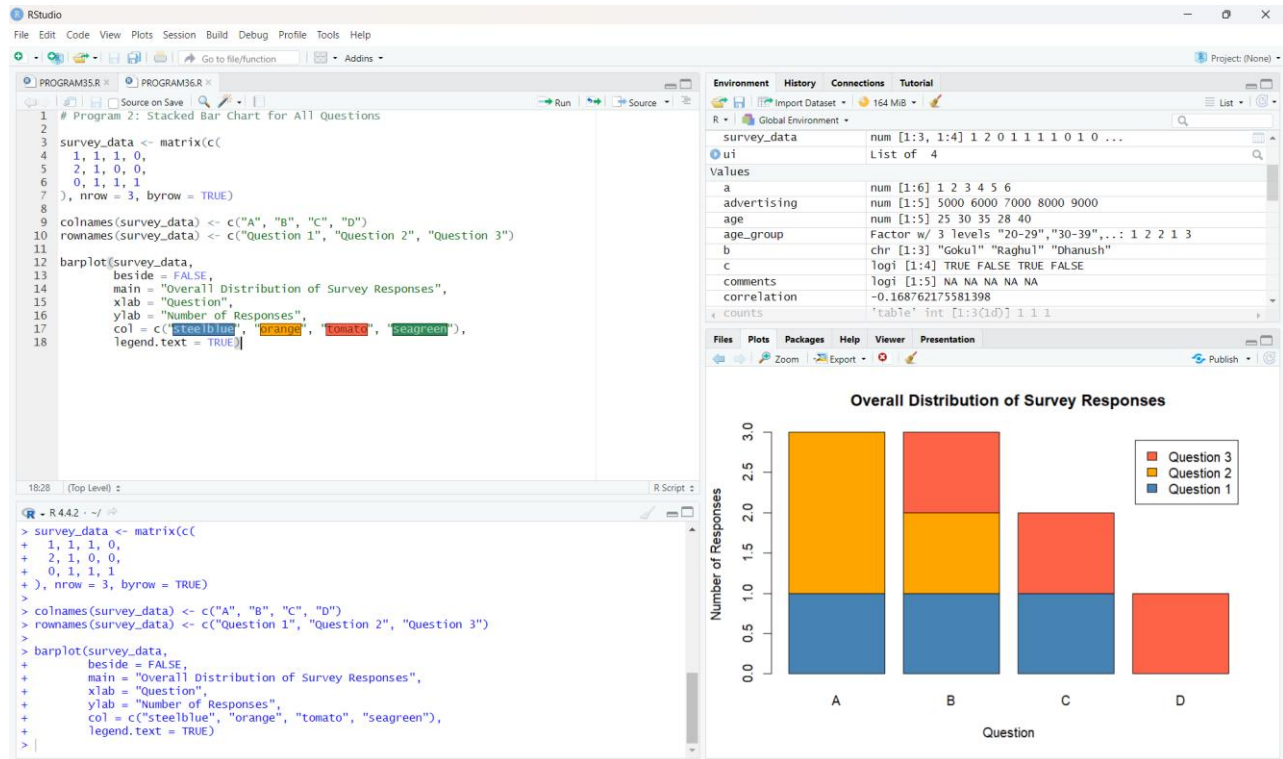
```
# Program 2: Stacked Bar Chart for All Questions

survey_data <- matrix(c(
  1, 1, 1, 0,
  2, 1, 0, 0,
  0, 1, 1, 1
), nrow = 3, byrow = TRUE)

colnames(survey_data) <- c("A", "B", "C", "D")
rownames(survey_data) <- c("Question 1", "Question 2", "Question 3")

barplot(survey_data,
  beside = FALSE,
  main = "Overall Distribution of Survey Responses",
  xlab = "Question",
  ylab = "Number of Responses",
  col = c("steelblue", "orange", "tomato", "seagreen"),
  legend.text = TRUE)
```

Output 2: Overall Distribution of Survey Responses



Output 2: Stacked Bar Chart for All Questions

Observation: The stacked bar chart displays the response distribution across answer categories A, B, C, and D for the three questions.

PROGRAM 3: Survey Response Table

Build a table to show the survey response data for each survey. Label the table elements.

```
# Program 3: Survey Response Table
```

```
survey_data <- data.frame(  
  Survey_ID = c(1, 2, 3),  
  Question_1 = c("A", "B", "C"),  
  Question_2 = c("B", "A", "A"),  
  Question_3 = c("C", "D", "B")  
)  
  
print(survey_data)
```

Output 3: Survey Response Table

The screenshot shows the RStudio interface. The script editor on the left contains the following R code:

```

1 # Program 3: Survey Response Table
2
3 survey_data <- data.frame(
4   Survey_ID = c(1, 2, 3),
5   Question_1 = c("A", "B", "C"),
6   Question_2 = c("B", "A", "A"),
7   Question_3 = c("C", "D", "B")
8 )
9
10 print(survey_data)

```

The Environment pane on the right shows the objects in the global environment:

- survey_data**: 3 obs. of 4 variables
- ui**: List of 4

The console at the bottom shows the execution of the code and the resulting output table:

```

R - R 4.4.2 ~ -/
+ col = c("steelblue", "orange", "tomato", "seagreen"),
+ legend.text = TRUE)
+ # Program 3: Survey Response Table
+
+ > survey_data <- data.frame(
+   Survey_ID = c(1, 2, 3),
+   Question_1 = c("A", "B", "C"),
+   Question_2 = c("B", "A", "A"),
+   Question_3 = c("C", "D", "B")
+ )
+
+ > print(survey_data)
  Survey_ID Question_1 Question_2 Question_3
1         1         A         B         C
2         2         B         A         D
3         3         C         A         B

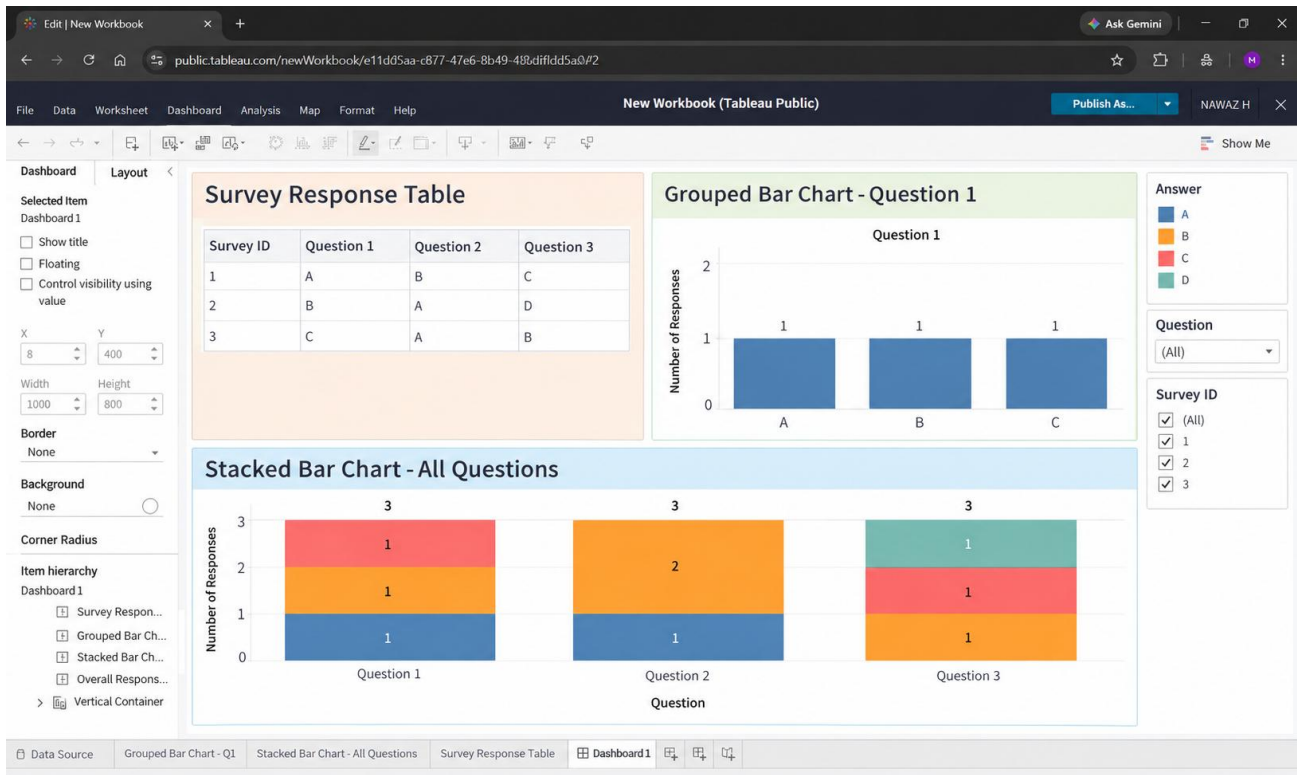
```

Output 3: Survey Response Table

Observation: The table displays the three survey records and their responses to Questions 1, 2, and 3.

PROGRAM 4: Tableau Dashboard

Develop a Tableau dashboard combining the grouped bar chart, stacked bar chart, and survey response table for interactive exploration of survey responses.



Output 4: Tableau Survey Responses Dashboard

Observation: The Tableau dashboard combines the survey response table, Question 1 bar chart, and stacked bar chart. The available filters allow the survey data to be explored interactively.

Procedure

1. Create the survey response vectors and data frame in R.
2. Use `table()` and `barplot()` to visualize the distribution of Question 1 responses.
3. Create the response matrix and use `barplot()` with `beside = FALSE` for the stacked bar chart.
4. Create and print the survey response table using `data.frame()`.
5. Enter the survey response data into Tableau and create the required worksheets.
6. Combine the worksheets into a dashboard and add filters for interactive exploration.

Result

The survey response data was successfully visualized using a bar chart, stacked bar chart, and table in R. A Tableau dashboard was also created by combining the three visualizations for interactive exploration.

Key R Functions Used

Function	Purpose
<code>table()</code>	Creates frequency counts of survey responses.
<code>barplot()</code>	Creates grouped and stacked bar charts.
<code>matrix()</code>	Creates the response matrix for the stacked chart.
<code>data.frame()</code>	Creates the survey response table.
<code>print()</code>	Displays the survey response data.